C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 8.01

Integration & Kinematics

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 D	Q3 D	Q4 C	Q5 B
Q6 C	Q7 D	Q8 D	Q9 A	Q10 A
Q11 D	Q12 D	Q13 C	Q14 B	Q15 C
Q16 B	Q17 A	Q18 C	Q19 A	Q20 A
Q21 D	Q22 A	Q23 C	Q24 D	Q25 A

Integration & Kinematics Practice Exercises

1 MCQ

Evaluate the definite integral: $\int_0^1 (2x+1)^3\,dx$.

A $\frac{80}{2}$

B $\frac{84}{8}$

C $\frac{80}{4}$

D $\frac{80}{8}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 2x+1 \implies du = 2\,dx$. Integral: $\frac{1}{2}\left[\frac{u^4}{4}\right]_1^3 = \frac{(3)^4-1^4}{8} = \frac{80}{8}$. [M1][A1]

2 MCQ

Evaluate the definite integral: $\int_0^1 (3x+1)^3\,dx$.

A $\frac{255}{4}$

B $\frac{255}{3}$

C $\frac{259}{12}$

D $\frac{255}{12}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 3x+1 \implies du = 3\,dx$. Integral: $\frac{1}{3}\left[\frac{u^4}{4}\right]_1^4 = \frac{(4)^4-1^4}{12} = \frac{255}{12}$. [M1][A1]

3 MCQ

Evaluate the definite integral: $\int_0^1 (4x+1)^3\,dx$.

A $\frac{625}{16}$

B $\frac{624}{4}$

C $\frac{628}{16}$

D $\frac{624}{16}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 4x+1 \implies du = 4\,dx$. Integral: $\frac{1}{4}\left[\frac{u^4}{4}\right]_1^5 = \frac{(5)^4-1^4}{16} = \frac{624}{16}$. [M1][A1]

4 MCQ

Evaluate the definite integral: $\int_0^1 (5x+1)^3\,dx$.

A $\frac{1295}{5}$

B $\frac{1295}{4}$

C $\frac{1295}{20}$ ✓

D $\frac{1299}{20}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 5x+1 \implies du = 5\,dx$. Integral: $\frac{1}{5}\left[\frac{u^4}{4}\right]_1^6 = \frac{(6)^4-1^4}{20} = \frac{1295}{20}$. [M1][A1]

5 MCQ

Evaluate the definite integral: $\int_0^1 (6x+1)^3\,dx$.

A $\frac{2400}{6}$

B $\frac{2400}{24}$ ✓

C $\frac{2404}{24}$

D $\frac{2400}{4}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 6x+1 \implies du = 6\,dx$. Integral: $\frac{1}{6}\left[\frac{u^4}{4}\right]_1^7 = \frac{(7)^4-1^4}{24} = \frac{2400}{24}$. [M1][A1]

6 MCQ

Evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

A $\frac{4095}{4}$

B $\frac{4099}{28}$

C $\frac{4095}{28}$ ✓

D $\frac{4095}{7}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 7x+1 \implies du = 7\,dx$. Integral: $\frac{1}{7}\left[\frac{u^4}{4}\right]_1^8 = \frac{(8)^4-1^4}{28} = \frac{4095}{28}$. [M1][A1]

7 MCQ

Evaluate the definite integral: $\int_0^1 (8x+1)^3\,dx$.

A $\frac{6560}{4}$

B $\frac{6560}{8}$

C $\frac{6564}{32}$

D $\frac{6560}{32}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 8x+1 \implies du = 8\,dx$. Integral: $\frac{1}{8}\left[\frac{u^4}{4}\right]_1^9 = \frac{(9)^4-1^4}{32} = \frac{6560}{32}$. [M1][A1]

8 MCQ

Evaluate the definite integral: $\int_0^1 (9x+1)^3\,dx$.

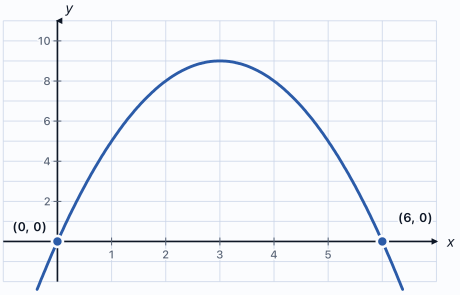
A $\frac{9999}{4}$	B $\frac{9999}{9}$
C $\frac{10003}{36}$	D $\frac{9999}{36}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 9x+1 \implies du = 9\,dx$. Integral: $\frac{1}{9}\left[\frac{u^4}{4}\right]_1^{10} = \frac{(10)^4-1^4}{36} = \frac{9999}{36}$. [M1][A1]

9 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 36 ✓	B 27
C 54	D 72

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$:

$$\text{Area} = \int_0^6 (6x - x^2)\,dx$$

Integrate term by term:

$$\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$$

Substitute the limits:

$$\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$$

Common errors:

- Estimating the region as a triangle with base 6 and maximum height 9 yields $\frac{1}{2} \times 6 \times 9 = 27$.
- Taking the bounding rectangle gives $6 \times 9 = 54$.
- Integrating $6x$ as $6x^2$ without dividing by 2 gives $216 - 72 = 144$, or subtracting $108 - 36 = 72$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 36. **[A1]**

10 MCQ

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant.\n\nGiven that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

- **A** 6 m ✓
- **B** 4 m
- **C** 5 m
- **D** 8 m

WORKED SOLUTION & VERIFICATION

Step 1: Determine the unknown parameter k.\nThe particle is momentarily at rest when $t = 2$, so $v(2) = 0$:\n

$$6(2)^2 - 18(2) + k = 0 \implies 24 - 36 + k = 0 \implies k = 12$$

Step 2: Find all times when the particle is at rest to check for direction changes.\n

$$v(t) = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2) = 6(t - 1)(t - 2) = 0$$

The particle is at rest at $t = 1$ and $t = 2$. In the interval $0 \le t < 1$, $v(t) > 0$; in the interval $1 < t < 2$, $v(t) < 0$.\n
Step 3: Calculate the total distance traveled by integrating over each interval.\nFor $0 \le t \le 1$:\n

$$s_1 = \int_0^1 (6t^2 - 18t + 12)\,dt = [2t^3 - 9t^2 + 12t]_0^1 = 2(1)^3 - 9(1)^2 + 12(1) - 0 = 5 \text{ m}$$

For $1 \le t \le 2$:\n

$$s_2 = \int_1^2 (6t^2 - 18t + 12)\,dt = [2t^3 - 9t^2 + 12t]_1^2 = (2(8) - 9(4) + 12(2)) - 5 = 4 - 5 = -1 \text{ m}$$

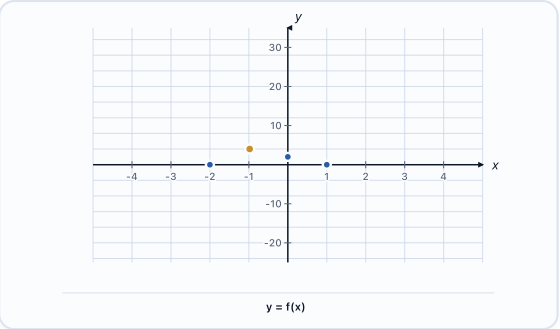
The distance traveled in this interval is $|-1| = 1$ m.\n
Step 4: Sum the distances.\n

$$\text{Total distance} = 5 + 1 = 6 \text{ m}$$

Common Trap: Evaluating simply the net displacement $\int_0^2 v\,dt = 4$ m neglects the reversal of direction at $t = 1$. [M1][A1]

11 MCQ

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.



A 84

B 64

C 12

D **108** ✓

WORKED SOLUTION & VERIFICATION

To find the area of the enclosed region, we synthesize differential calculus, polynomial algebra, and definite integration:

1. **Find the equation of the tangent line at $P(2, 4)$:**

Differentiating the curve equation:

$$\frac{dy}{dx} = 3x^2 - 3$$

At $x = 2$, the gradient of the tangent is:

$$m = 3(2)^2 - 3 = 12 - 3 = 9$$

Using the point-slope formula with $P(2, 4)$:

$$y - 4 = 9(x - 2) \implies y = 9x - 14$$

2. **Find the intersection point Q:**

Set the curve and tangent equations equal:

$$x^3 - 3x + 2 = 9x - 14 \implies x^3 - 12x + 16 = 0$$

Since the line is tangent to the curve at $x = 2$, $(x - 2)^2$ must be a factor of the polynomial:

$$(x - 2)^2 = x^2 - 4x + 4$$

Dividing $x^3 - 12x + 16$ by $x^2 - 4x + 4$ gives $(x + 4)$:

$$(x - 2)^2(x + 4) = 0$$

Thus, the other intersection point Q occurs at $x = -4$.

3. **Compute the enclosed area:**

For $x \in (-4, 2)$, the curve lies above the tangent line (e.g., at $x = 0$, $y_{\text{curve}} = 2$ while $y_{\text{tangent}} = -14$).

The enclosed area A is given by:

$$A = \int_{-4}^{2} [(x^3 - 3x + 2) - (9x - 14)]\, dx = \int_{-4}^{2} (x^3 - 12x + 16)\, dx$$

Finding the antiderivative:

$$F(x) = \frac{x^4}{4} - 6x^2 + 16x$$

Evaluating at the boundaries:

$$F(2) = \frac{2^4}{4} - 6(2)^2 + 16(2) = 4 - 24 + 32 = 12$$

$$F(-4) = \frac{(-4)^4}{4} - 6(-4)^2 + 16(-4) = 64 - 96 - 64 = -96$$

Therefore:

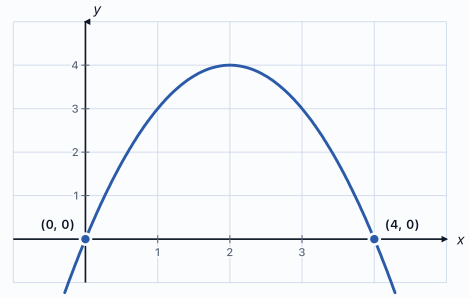
$$A = F(2) - F(-4) = 12 - (-96) = 108$$

Common Misconceptions:

- Choice B (84) results from an arithmetic sign error when expanding $F(-4)$ as $64 + 96 - 64 = 96$, leading to $12 - (-96) \rightarrow 96 - 12 = 84$.
- Choice C (64) arises from incorrectly assuming the region is symmetric about the y-axis and integrating over $[-2, 2]$.
- Choice D (12) arises from mistakenly integrating only over the positive domain from the y-axis $[0, 2]$. [M1][A1]

12 MCQ

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



A 16

B $\frac{64}{3}$

C $\frac{16}{3}$

D $\frac{32}{3}$ ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** To find the area enclosed by the curve and the x-axis, first find the x-intercepts by setting $y = 0$:

$$4x - x^2 = 0 \implies x(4 - x) = 0 \implies x = 0 \text{ or } x = 4.$$

The area is given by the definite integral of y with respect to x from $x = 0$ to $x = 4$:

$$\int_0^4 (4x - x^2)\,dx = [2x^2 - \frac{x^3}{3}]_0^4$$

Evaluating between the limits:

$$(2(4)^2 - \frac{4^3}{3}) - 0 = 32 - \frac{64}{3} = \frac{96 - 64}{3} = \frac{32}{3}.$$

Common misconceptions include:

- Integrating without the divisor 3 to get $32 - 16 = 16$.
- Doubling the integral mistakenly to get $\frac{64}{3}$.
- Integrating only up to the vertex at $x = 2$ to get $\frac{16}{3}$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: $\frac{32}{3}$. **[A1]**

13 MCQ

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

A -25

B -13

C -37 ✓

D -5

WORKED SOLUTION & VERIFICATION

1. **Find the velocity function $v(t)$:**

$$v(t) = \int a\,dt = \int (6t - 18)\,dt = 3t^2 - 18t + c$$

Since the particle is at instantaneous rest when $t = 1$, $v(1) = 0$:

$$3(1)^2 - 18(1) + c = 0 \implies -15 + c = 0 \implies c = 15$$

Thus, $v(t) = 3t^2 - 18t + 15$.

2. **Find when the particle is next at instantaneous rest:**

Set $v(t) = 0$:

$$3(t^2 - 6t + 5) = 0 \implies 3(t - 1)(t - 5) = 0$$

So the particle is at rest at $t = 1$ and next at $t = 5$.

3. **Find the displacement function $s(t)$:**

$$s(t) = \int v(t)\,dt = \int (3t^2 - 18t + 15)\,dt = t^3 - 9t^2 + 15t + d$$

Given that $s(2) = -10$:

$$(2)^3 - 9(2)^2 + 15(2) + d = -10$$

$$8 - 36 + 30 + d = -10 \implies 2 + d = -10 \implies d = -12$$

Thus, $s(t) = t^3 - 9t^2 + 15t - 12$.

4. **Evaluate displacement at $t = 5$:**

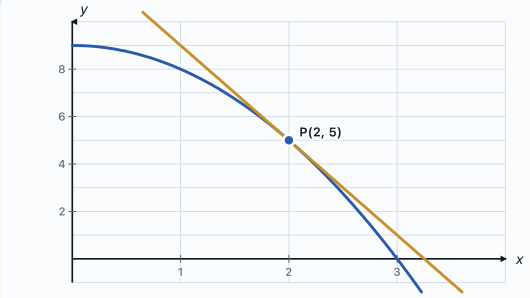
$$s(5) = (5)^3 - 9(5)^2 + 15(5) - 12 = 125 - 225 + 75 - 12 = -37 \text{ m}$$

Common traps:

- Setting $d = 0$ (assuming displacement is zero at $t = 0$) leads to $s(5) = -25$.
- A sign error when solving $2 + d = -10$ yielding $d = 12$ leads to $s(5) = -13$.
- Evaluating at the initial rest time $t = 1$ instead of the next rest time gives $s(1) = -5$. [M1][A1]

14 MCQ

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.



A $\frac{125}{192}$

B $\frac{11}{24}$ ✓

C $\frac{8}{3}$

D $\frac{25}{8}$

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**

- **Step 2 (Derivation & Calculation):** 1. **Find the equation of the tangent at $P(2, 5)$:**

Given $y = 9 - x^2$, differentiating with respect to x gives:

$$\frac{dy}{dx} = -2x$$

At $x = 2$, the gradient of the tangent is $m = -2(2) = -4$.

The equation of the tangent line is:

$$y - 5 = -4(x - 2) \implies y = 13 - 4x$$

2. **Determine the x-intercepts:**

- Tangent: Set $y = 0 \implies 13 - 4x = 0 \implies x = \frac{13}{4} = 3.25$.

- Curve: Set $y = 0 \implies 9 - x^2 = 0 \implies x = 3$ (since $x \geq 0$).

3. **Calculate the area of the bounded region:**

The region lies between $x = 2$ and $x = \frac{13}{4}$. We can calculate the area by splitting it into two sections along the x-axis:

- From $x = 2$ to $x = 3$, the region is bounded above by the tangent and below by the curve:

$$A_1 = \int_2^3 [(13 - 4x) - (9 - x^2)]\, dx = \int_2^3 (x - 2)^2\, dx = [\frac{(x-2)^3}{3}]_2^3 = \frac{1}{3} - 0 = \frac{1}{3}$$

- From $x = 3$ to $x = \frac{13}{4}$, the region is bounded above by the tangent and below by the x-axis ($y = 0$):

$$A_2 = \int_3^{13/4} (13 - 4x)\, dx = [13x - 2x^2]_3^{13/4} = (13\,(\frac{13}{4}) - 2\,(\frac{169}{16})) - (39 - 18) = \frac{169}{8} - 21 = \frac{1}{8}$$

Adding the two areas:

$$\text{Area} = A_1 + A_2 = \frac{1}{3} + \frac{1}{8} = \frac{11}{24}$$

(Alternatively, subtract the area under the curve from $x = 2$ to 3 from the right-angled triangle under the tangent line from $x = 2$ to $x = 3.25$: $\frac{1}{2} \times \frac{5}{4} \times 5 - \int_2^3 (9 - x^2)\, dx = \frac{25}{8} - \frac{8}{3} = \frac{11}{24}$.)

Common Misconceptions:

- $\frac{125}{192}$ occurs from blindly evaluating $\int_2^{13/4} (x - 2)^2\, dx$, incorrectly assuming the curve remains the lower boundary

past its x-intercept at $x = 3$.
- $\frac{8}{3}$ is the area bounded by the curve, the tangent, and the y-axis instead of the x-axis.
- $\frac{25}{8}$ is the area of the triangle under the tangent without subtracting the region under the curve. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: $\frac{11}{24}$. **[A1]**

15 MCQ

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is 5 m s^{-1}. Find the velocity of the particle when $t = 3$.

A 19 m s^{-1}

B 27 m s^{-1}

C 21 m s^{-1} ✓

D 15 m s^{-1}

WORKED SOLUTION & VERIFICATION

To find the velocity v, integrate the acceleration function with respect to t:

$$v = \int a\,\mathrm{d}t = \int (6t - 4)\,\mathrm{d}t = 3t^2 - 4t + C$$

Using the initial condition that $v = 5$ when $t = 1$:

$$3(1)^2 - 4(1) + C = 5 \implies -1 + C = 5 \implies C = 6$$

Thus, the velocity function is:

$$v(t) = 3t^2 - 4t + 6$$

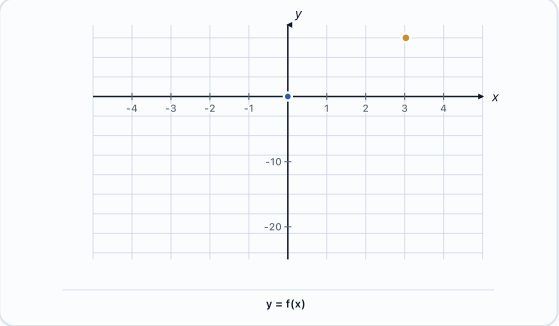
Substitute $t = 3$:

$$v(3) = 3(3)^2 - 4(3) + 6 = 27 - 12 + 6 = 21 \text{ m s}^{-1}$$

Common misconceptions:
- Omitting the constant of integration C gives $3(3)^2 - 4(3) = 15$ m s^{-1}.
- A sign error when solving $3 - 4 + C = 5$ (such as writing $1 + C = 5 \implies C = 4$) leads to 19 m s^{-1}. [M1][A1]

16 MCQ

A curve has equation $y = 6x - x^2$. The straight line $y = mx$, where $0 < m < 6$, intersects the curve at the origin and at the point P. Given that the area of the region enclosed between the curve and the line is $\frac{32}{3}$, find the value of the constant m.



A 4

B 2 ✓

C 3

D 1

WORKED SOLUTION & VERIFICATION

Step 1: Find the points of intersection of the curve and the line.

Set the equations equal to each other:

$$6x - x^2 = mx \implies x^2 - (6-m)x = 0 \implies x(x-(6-m)) = 0$$

Since the line intersects the curve at the origin $(0, 0)$ and at point P, the x-coordinate of P is $x = 6 - m$.

Step 2: Set up the definite integral for the enclosed area.

For $0 \le x \le 6 - m$, the curve lies above the straight line. The enclosed area A is given by:

$$A = \int_0^{6-m} ((6x - x^2) - mx)\,\mathrm{d}x = \int_0^{6-m} ((6-m)x - x^2)\,\mathrm{d}x$$

Step 3: Integrate and evaluate at the limits.

$$A = \left[\frac{6-m}{2}x^2 - \frac{x^3}{3}\right]_0^{6-m} = \frac{(6-m)(6-m)^2}{2} - \frac{(6-m)^3}{3} = \frac{(6-m)^3}{6}$$

Step 4: Solve for m.

We are given $A = \frac{32}{3}$:

$$\frac{(6-m)^3}{6} = \frac{32}{3} \implies (6-m)^3 = 64 \implies 6 - m = 4 \implies m = 2$$

Common Pitfall:

A common error is to confuse the x-coordinate of the intersection point P (which is 4) with the value of the parameter m, leading to the incorrect choice 4. [M1][A1]

17 MCQ

Calculate the solution set for the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

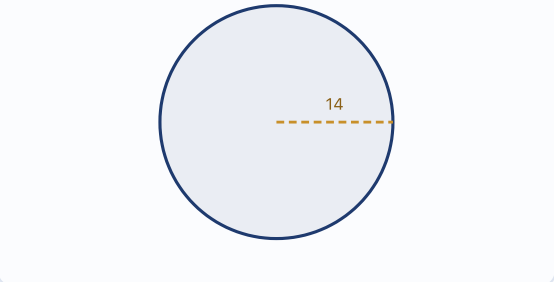
- **A** $x = \frac{4 \ln 2}{\ln 5 - \ln 2}$ ✓
- **B** $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$
- **C** $x = \frac{4 \ln 5}{\ln 5 - \ln 2}$
- **D** $x = \frac{4}{\ln 5 - \ln 2}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x+4)\ln 2 = x \ln 5 \implies x \ln 2 + 4 \ln 2 = x \ln 5 \implies x(\ln 5 - \ln 2) = 4 \ln 2 \implies x = \frac{4 \ln 2}{\ln 5 - \ln 2}$. [M1][A1]

18 MCQ

Suppose a sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?



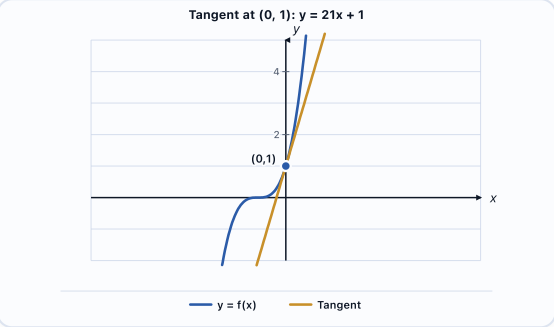
- **A** 16.8 cm^2
- **B** 235.2 cm^2
- **C** 117.6 cm^2 ✓
- **D** 122.6 cm^2

WORKED SOLUTION & VERIFICATION

Sector area formula: $A = \frac{1}{2} r^2 \theta = \frac{1}{2}(14)^2(1.2) = 117.6 \text{ cm}^2$. [M1][A1]

19 MCQ

Identify the equation of the tangent line to the curve $y = (7x+1)^3$ at the point where $x = 0$.



- **A** $y = 21x + 1$ ✓
- **B** $y = 21x + 3$
- **C** $y = 21x - 1$
- **D** $y = 7x + 1$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(7x+1)^2 \cdot (7) = 21(7x+1)^2$. At $x = 0$, slope $m = 21$. Equation: $y - 1 = 21(x - 0) \implies y = 21x + 1$. [M1][A1]

20 MCQ

Determine the numerical value of the definite integral: $\int_0^1 (3x+1)^3\,dx$.

- **A** $\frac{255}{12}$ ✓
- **B** $\frac{255}{4}$
- **C** $\frac{255}{3}$
- **D** $\frac{259}{12}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 3x+1 \implies du = 3\,dx$. Integral: $\frac{1}{3}\left[\frac{u^4}{4}\right]_1^4 = \frac{(4)^4-1^4}{12} = \frac{255}{12}$. [M1][A1]

21 MCQ

Calculate the simplified result of the definite integral: $\int_0^1 (4x+1)^3\,dx$.

- **A** $\frac{625}{16}$
- **B** $\frac{624}{4}$
- **C** $\frac{628}{16}$
- **D** $\frac{624}{16}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 4x+1 \implies du = 4\,dx$. Integral: $\frac{1}{4}\left[\frac{u^4}{4}\right]_1^5 = \frac{(5)^4-1^4}{16} = \frac{624}{16}$. [M1][A1]

22 MCQ

Find the exact value of the expression the definite integral: $\int_0^1 (5x+1)^3\,dx$.

- **A** $\frac{1295}{20}$ ✓
- **B** $\frac{1295}{5}$
- **C** $\frac{1295}{4}$
- **D** $\frac{1299}{20}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 5x+1 \implies du = 5\,dx$. Integral: $\frac{1}{5}\left[\frac{u^4}{4}\right]_1^6 = \frac{(6)^4-1^4}{20} = \frac{1295}{20}$. [M1][A1]

23 MCQ

Work out the exact value of the definite integral: $\int_0^1 (6x+1)^3\,dx$.

- **A** $\frac{2404}{24}$
- **B** $\frac{2400}{4}$
- **C** $\frac{2400}{24}$ ✓
- **D** $\frac{2400}{6}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 6x+1 \implies du = 6\,dx$. Integral: $\frac{1}{6}\left[\frac{u^4}{4}\right]_1^7 = \frac{(7)^4-1^4}{24} = \frac{2400}{24}$. [M1][A1]

24 MCQ

Simplify and evaluate the definite integral: $\int_0^1 (7x+1)^3\,dx$.

A $\frac{4095}{4}$

B $\frac{4099}{28}$

C $\frac{4095}{7}$

D $\frac{4095}{28}$ ✓

WORKED SOLUTION & VERIFICATION

Substitute $u = 7x+1 \implies du = 7\,dx$. Integral: $\frac{1}{7}\left[\frac{u^4}{4}\right]_1^8 = \frac{(8)^4-1^4}{28} = \frac{4095}{28}$. [M1][A1]

25 MCQ

Ascertain the exact result of the definite integral: $\int_0^1 (8x+1)^3\,dx$.

A $\frac{6560}{32}$ ✓

B $\frac{6560}{4}$

C $\frac{6560}{8}$

D $\frac{6564}{32}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 8x+1 \implies du = 8\,dx$. Integral: $\frac{1}{8}\left[\frac{u^4}{4}\right]_1^9 = \frac{(9)^4-1^4}{32} = \frac{6560}{32}$. [M1][A1]